SOLAR PLANT GALASCOPE 2.5 MW

ORGANIC ANALYSIS – SOFTWARE & HARDWARE CONFIGURATION SYSTEM

REVIEWS	**MODIFICATIONS**	**DATES**	**INITIALS AND VISAS**		
			DRAFT	REVIEW	APPROVAL
0	As Built	24/04/20	JPM		
1	After Site modification	24/04/21	JPM		
2					
3					
4					
5					
6					
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9					
10					
11					

SUMMARY

1 INTRODUCTION

This document describes the overall structure of common treatments, and software identification of Solar Plant level 1's software. It also describes the several operations on software handling and hardware configuration.

2 SYSTEM OVER VIEW

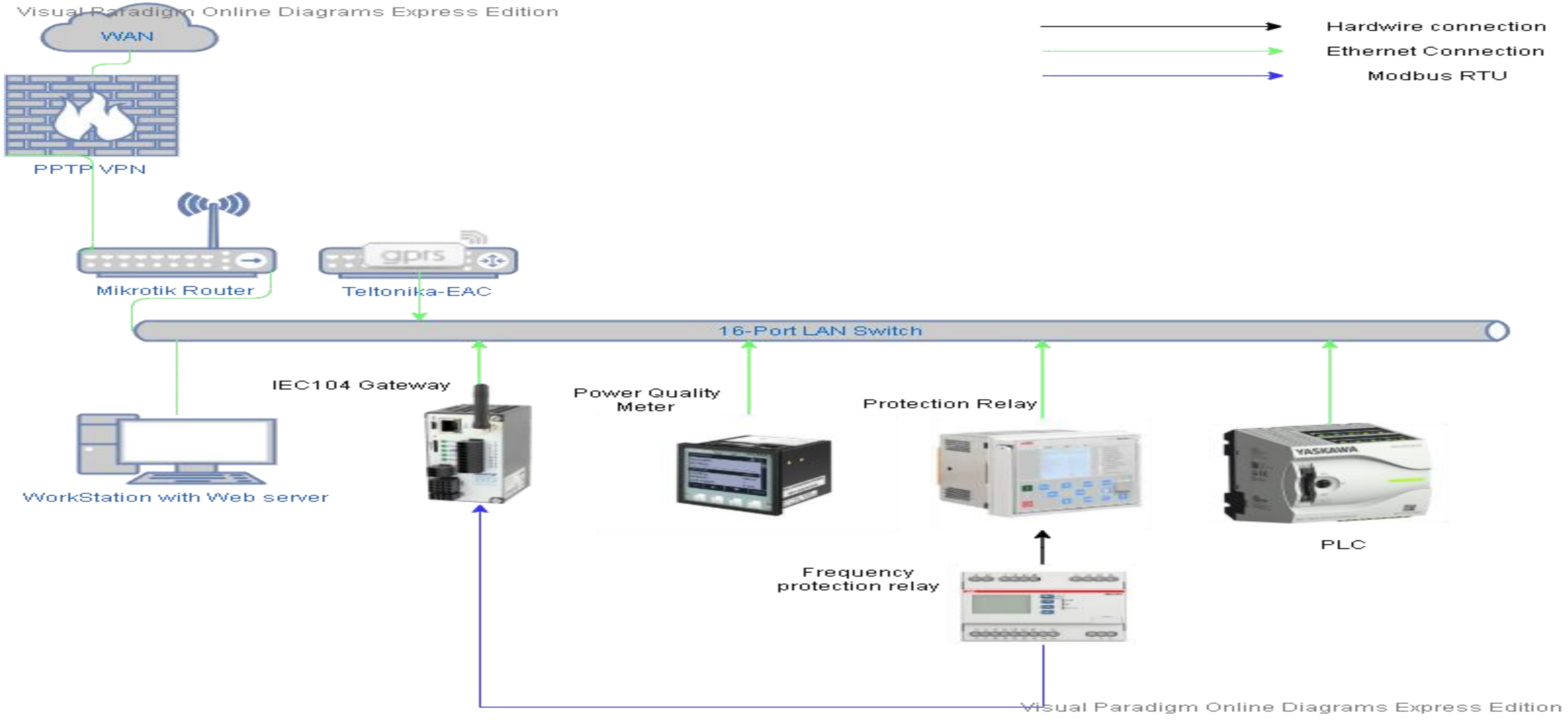
This document describes the overall organic structure of Solar Plant level 1's software which is programmed with Simatic, , web interface.

2.1 Abbreviations

List of abbreviations and codes used in this document:

- PLC : Programmable Logic Controller
- PC : Personal Computer
- SCADA : Human Machine Interface

3 ARCHITECTURE



4 PLC OVERALL SOFTWARE PROGRAMMES

The main controls program file MAIN is used to call subroutines, secondary program files and generally sequences the major activities within the software system.

Subroutines are used when a function is performed in more than one place in the application program or when specific application program is performed in the PLC.

4.1 The structure of the program.

BLOCKS NUMBER	DESIGNATION
OB 1	MAIN PROGRAM
OB 100	Complete Restart
OB 101	Restart
OB 102	Cold Restart
FB 63	Standard Send Function
FB 64	Standard Receive Function
FB 65	Standard Connection Function
FB 66	Standard Disconnection Function
FB 71	Standard Modbus Servers Function
FC 2	Power Management
FC 97	Set TCP endpoint
UDT	Structure of connection parameter

4.2 DATA MAPPING

	TYPE		DESIGNATION
QB12 to QB15	OUTPUT	Q	PHYSICAL DIGITAL OUTPUT
IB00 to IB11	INPUT	I	PHYSICAL DIGITAL INPUT
COMMON DATA			
MB0	BITS	M	PLC CLOCK MEMORY (PULSEPAUSE)
MB1	BITS	M	PLC CLOCK MEMORY (PULSEPAUSE) RISING EDGE
MB2	BITS	M	TIME PULSE
M4.0	BIT	M	FIRST SCAN BIT
M4.1	BIT	M	BIT ALWAYS ON
M4.0	BIT	M	BIT ALWAYS OFF
T001 to T254	TIMER	T	MISCELLANEOUS TIMERS
Z001 to Z254	COUNTER	Z	MISCELLANEOUS COUNTERS
DATA BLOCK AFFECTATION			
DB1	DATA BLOCK	DB	COMMUNICATION PARAMETER
DB2	DATA BLOCK	DB	PULSE ARRAY
DB10	DATA BLOCK	DB	EXCHANGE TABLE BETWEEN PLC AND GATEWAY
DB65	DATA BLOCK	DB	INSTANCE BLOCK OF TCON FUNCTION
DB71	DATA BLOCK	DB	INSTANCE BLOCK OF MODBUS SERVER FUNCTION
DB72	DATA BLOCK	DB	IMAGE OF DB 71
UDT64	STRUCTURE	DB	STRUCTURE FOR COMMUNICATION CONNECTION APPLY IN DB1

DATA BLOCK COMMUNICATION STRUCTURE:

Address	Name	Type	Initial Value	Description
0	block_length	WORD	W#16#40	Length of the UDT (64 bytes)
2	id	WORD	W#16#1	Reference to this connection (valuerange: W#16#0001 to W#16#FFFF)
4	connection_type	BYTE	B#16#11	B#16#11: TCP/IP native; B#16#12: ISO on TCP; B#16#13: UDP; B#16#01: TCP/IP comp
5	active_est	BOOL	FALSE	FALSE: passive connection establishment; TRUE: active connection establishment
6	local_device_id	BYTE	B#16#2	Allowed values: B#16#0, B#16#2, B#16#3, B#16#5. See online help.
7	local_tsap_id_len	BYTE	B#16#2	Used length of the parameter local_tsap_id
8	rem_subnet_id_len	BYTE	B#16#0	Unused; must be B#16#00
9	rem_staddr_len	BYTE	B#16#0	Meaning of parameter rem_staddr: B#16#00: is irrelevant; B#16#04: valid address
10	rem_tsap_id_len	BYTE	B#16#0	Used length of the parameter rem_tsap_id
11	next_staddr_len	BYTE	B#16#0	B#16#1 if local_device_id = 0; else B#16#0
12	local_tsap_id	ARRAY[1..16]	B#16#7, B#16#D0	Depending on parameter connection_type: local port no. / local TSAP-ID
		BYTE		
28	rem_subnet_id	ARRAY[1..6]	B#16#0	Unused; must be B#16#00
		BYTE		
34	rem_staddr	ARRAY[1..6]	B#16#0	IP address of the remote connection end point, e. g. 192.168.0.1
		BYTE		
40	rem_tsap_id	ARRAY[1..16]	B#16#0	Depending on connection type: remote port no. / remote TSAP-ID
		BYTE		
56	next_staddr	ARRAY[1..6]	B#16#0	Depending on local_device_id: rack / slot no. of the associated CP / irrelevant
		BYTE		
62	spare	WORD	W#16#0	Unused; must be W#16#0000

DATA BLOCK EXCCHANGES

Sent by Gateway		
Rank	Byte	Description
0	0	SP Active Power
1	2	SP Reactive Power
2	4	SP Active Power 100%
3	6	SP Active Power 60%
4	8	SP Active Power 30%
5	10	SP Active Power 0%
6	12	Demand Active Power SP
7	14	Demand Reactive Power SP
8	16	Circuit Breaker Control
9	18	Circuit Breaker Remote/Local
10	20	Com Status PR
11	22	Com Status Inverter
12	24	Circuit Breaker Open/Close
13	26	Actual Active Power
Read by Gateway		
Rank	Byte	Description
15	30	FB Active Power
16	32	FB Reactive Power
17	34	FB Active Power 100%
18	36	FB Active Power 60%
19	38	FB Active Power 30%
20	40	FB Active Power 0%
21	42	Active Power SP Activated
22	44	Reactive Power SP Activated
23	46	Circuit Breaker Open/Close
24	48	Circuit Breaker Remote/Local
25	50	Active Power Set point new value (Accepted/Rejected)
26	52	Reactive Power Set point new value (Accepted/Rejected)
27	54	CB Control
28	56	Active set point send to inverter
29	58	Reactive set point send to inverter

5 GATEWAY OVERALL SOFTWARE PROGRAMMES

The IEC 104 Gateway is used as server. It is in charge to collect all data from all devices make some basic computing send Set point to the Smart logger and send /receive command and status to the DSO scada.

Here under you will find all the data exchanges. At the annex 1 you will find DSO requirements'

5.1 Data Exchange Modbus.

PLC

Inputs	**tcp:10.2.2.1:502**	**Index**
MB_Active Power Setpoint Feedback	Read Holding Registers (0x03), start address: 15, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	0
MB_Reactive Power Setpoint Feedback	Read Holding Registers (0x03), start address: 16, quantity: 1, data type: 16-bit signed integer, unit identifier: 1	1
MB_ActivePower_Accepted	Read Holding Registers (0x03), start address: 21, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	2
MB_ReactivePower_Accepted	Read Holding Registers (0x03), start address: 22, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	3
MB_OpenClose_Cmd	Read Holding Registers (0x03), start address: 23, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	4
MB_CB_State	Read Holding Registers (0x03), start address: 24, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	5
MB_Active Power Setpoint 100 % Feedback	Read Holding Registers (0x03), start address: 17, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	6
MB_Active Power Setpoint 60 % Feedback	Read Holding Registers (0x03), start address: 18, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	7

MB_Active Power Setpoint 30 % Feedback	Read Holding Registers (0x03), start address: 19, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	8
MB_Active Power Setpoint 0 % Feedback	Read Holding Registers (0x03), start address: 20, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	9
MB_Active Power Setpoint new value (Accepted/Rejected)	Read Holding Registers (0x03), start address: 25, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	10
MB_Reactive Power Setpoint new value (Accepted/Rejected)	Read Holding Registers (0x03), start address: 26, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	11
MB_Active set point_send to inverter	Read Holding Registers (0x03), start address: 28, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	12
MB_Reactive set point_send to inverter	Read Holding Registers (0x03), start address: 29, quantity: 1, data type: 16-bit signed integer, unit identifier: 1	13
Outputs		
MB_Active Power Setpoint	Preset Multiple Registers (0x10), start address: 0, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	0
MB_Reactive Power Setpoint	Preset Multiple Registers (0x10), start address: 1, quantity: 1, data type: 16-bit signed integer, unit identifier: 1	1
MB_Active_Power_100%	Preset Multiple Registers (0x10), start address: 2, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	2
MB_Active_Power_60%	Preset Multiple Registers (0x10), start address: 3, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	3

MB_Active_Power_30%	Preset Multiple Registers (0x10), start address: 4, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	4
MB_Active_Power_0%	Preset Multiple Registers (0x10), start address: 5, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	5
MB_Demande_Active_Power	Preset Multiple Registers (0x10), start address: 6, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	6
MB_Demande_Reactive_Power	Preset Multiple Registers (0x10), start address: 7, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	7
MB_Circuit_Breaker_Control	Preset Multiple Registers (0x10), start address: 8, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	8
MB_Circuit_Breaker_RemoteLocal	Preset Multiple Registers (0x10), start address: 9, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	9
MB_Com_state_PR	Preset Multiple Registers (0x10), start address: 10, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	10
MB_Com_state_Inverter	Preset Multiple Registers (0x10), start address: 11, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	11
MB_Current ValueActive	Preset Multiple Registers (0x10), start address: 13, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	12

Power Quality Meter

Inputs	tcp:10.2.2.11:502	Index
PQM_PhVphA	Read Holding Registers (0x03), start address: 200, quantity: 2, data type: 32-bit floating point, unit identifier: 1	0
PQM_PhVphB	Read Holding Registers (0x03), start address: 202, quantity: 2, data type: 32-bit floating point, unit identifier: 1	1
PQM_PhVphC	Read Holding Registers (0x03), start address: 204, quantity: 2, data type: 32-bit floating point, unit identifier: 1	2
PQM_PPVphAB	Read Holding Registers (0x03), start address: 216, quantity: 2, data type: 32-bit floating point, unit identifier: 1	3
PQM_PPVphBC	Read Holding Registers (0x03), start address: 218, quantity: 2, data type: 32-bit floating point, unit identifier: 1	4
PQM_PPVphCA	Read Holding Registers (0x03), start address: 220, quantity: 2, data type: 32-bit floating point, unit identifier: 1	5
PQM_AphA	Read Holding Registers (0x03), start address: 208, quantity: 2, data type: 32-bit floating point, unit identifier: 1	6
PQM_AphB	Read Holding Registers (0x03), start address: 210, quantity: 2, data type: 32-bit floating point, unit identifier: 1	7
PQM_AphC	Read Holding Registers (0x03), start address: 212, quantity: 2, data type: 32-bit floating point, unit identifier: 1	8
PQM_w	Read Holding Registers (0x03), start address: 232, quantity: 2, data type: 32-bit floating point, unit identifier: 1	9
PQM_var	Read Holding Registers (0x03), start address: 240, quantity: 2, data type: 32-bit floating point, unit identifier: 1	10
PQM_VA	Read Holding Registers (0x03), start address: 248, quantity: 2, data type: 32-bit floating point, unit identifier: 1	11
PQM_PF	Read Holding Registers (0x03), start address: 264, quantity: 2, data type: 32-bit floating point, unit identifier: 1	12
PQM_HZ	Read Holding Registers (0x03), start address: 274, quantity: 2, data type: 32-bit floating point, unit identifier: 1	13

Protection relay

Inputs	**tcp:10.2.2.12:502**	**Index**
REL_LEDS_STATUS	Read Holding Registers (0x03), start address: 172, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	0
REL_175_STATUS	Read Holding Registers (0x03), start address: 174, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	1
REL_170_STATUS	Read Holding Registers (0x03), start address: 169, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	2
REL_250_STATUS	Read Holding Registers (0x03), start address: 249, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	3
REL_193_STATUS	Read Holding Registers (0x03), start address: 192, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	4
REL_196_STATUS	Read Holding Registers (0x03), start address: 195, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	5
REL_198_STATUS	Read Holding Registers (0x03), start address: 197, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	6
REL_182_STATUS	Read Holding Registers (0x03), start address: 181, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	7
REL_183_STATUS	Read Holding Registers (0x03), start address: 182, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	8
REL_189_STATUS	Read Holding Registers (0x03), start address: 188, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	9
REL_188_STATUS	Read Holding Registers (0x03), start address: 187, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	10
REL_129_STATUS	Read Holding Registers (0x03), start address: 128, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	11
Outputs		
Breaker_Cntrl_Open	Force Multiple Coils (0x0F), start address: 2051, quantity: 1, data type: Boolean, unit identifier: 1	0
Breaker_Cntrl_Close	Force Multiple Coils (0x0F), start address: 2052, quantity: 1, data type: Boolean, unit identifier: 1	1
Reset Protection	Force Multiple Coils (0x0F), start address: 2060, quantity: 1, data type: Boolean, unit identifier: 1	2

Smart Logger

Inputs	**tcp:10.2.2.10:502**	**Index**
ActivePower	Read Holding Registers (0x03), start address: 40525, quantity: 2, data type: 32-bit signed integer, unit identifier: 1	0
PowerFactor	Read Holding Registers (0x03), start address: 40532, quantity: 1, data type: 16-bit signed integer, unit identifier: 1	1
ReactivePower	Read Holding Registers (0x03), start address: 40544, quantity: 2, data type: 32-bit signed integer, unit identifier: 1	2
ETotal	Read Holding Registers (0x03), start address: 40560, quantity: 2, data type: 32-bit unsigned integer, unit identifier: 1	3
EDaily	Read Holding Registers (0x03), start address: 40564, quantity: 2, data type: 32-bit unsigned integer, unit identifier: 1	4
PlantStatus	Read Holding Registers (0x03), start address: 40566, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	5
InvertersEfficiency	Read Holding Registers (0x03), start address: 40685, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 1	6
WindSpeed	Read Holding Registers (0x03), start address: 40031, quantity: 1, data type: 16-bit signed integer, unit identifier: 59	7
PVTemp	Read Holding Registers (0x03), start address: 40033, quantity: 1, data type: 16-bit signed integer, unit identifier: 59	8
AmbientTemp	Read Holding Registers (0x03), start address: 40034, quantity: 1, data type: 16-bit signed integer, unit identifier: 58	9
SUnRadiation	Read Holding Registers (0x03), start address: 40035, quantity: 1, data type: 16-bit signed integer, unit identifier: 59	10
Inv01	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 4	11
Inv02	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 5	12
Inv03	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 6	13
Inv04	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 7	14
Inv05	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 8	15
Inv06	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 9	16
Inv07	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 10	17
Inv08	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 11	18

Inv09	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 12	19
Inv10	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 13	20
Inv11	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 14	21
Inv12	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 15	22
Inv13	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 16	23
Inv14	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 17	24
Inv15	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 18	25
Inv16	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 19	26
Inv17	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 20	27
Inv18	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 21	28
Inv19	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 22	29
Inv20	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 23	30
Inv21	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 24	31
Inv22	Read Holding Registers (0x03), start address: 32000, quantity: 1, data type: 16-bit unsigned integer, unit identifier: 25	32
Outputs		
ActiveAdjustement	Preset Multiple Registers (0x10), start address: 40424, quantity: 2, data type: 32-bit unsigned integer, unit identifier: 1	0
ReactiveAdjustement	Preset Multiple Registers (0x10), start address: 40426, quantity: 2, data type: 32-bit signed integer, unit identifier: 1	1

CM-UFD (FREQUENCY RELAY)

Inputs	**RS485 RTU COM0 ADDRESS 1 - 19200/EVEN/**	**Index**
UF1	Read Coil Status (0x01), start address: 32, quantity: 1, data type: Boolean	0
OF1	Read Coil Status (0x01), start address: 30, quantity: 1, data type: Boolean	1
ROCOF	Read Coil Status (0x01), start address: 34, quantity: 1, data type: Boolean	2

5.2 Data Exchange IEC104 (Server – ASDU : 115).

Inputs						
	Type ID	**IOA**	**Range**	**Interrogation Group**	**Cyclic Transmission Time**	**Index**
Active Power Setpoint (MW)	Set point command, short floating point value (50)	30	1	Not used (0)	0	0
Reactive Power Setpoint (MVar)	Set point command, short floating point value (50)	31	1	Not used (0)	0	1
Main Automatic Circuit breaker (Open/Close)	Double command (46)	10	1	Not used (0)	0	2
Active Power Setpoint 100% (On/Off)	Double command (46)	21	1	Not used (0)	0	3
Active Power Setpoint 60% (On/Off)	Double command (46)	22	1	Not used (0)	0	4
Active Power Setpoint 30% (On/Off)	Double command (46)	23	1	Not used (0)	0	5
Active Power Setpoint 0% (On/Off)	Double command (46)	24	1	Not used (0)	0	6
SCADA Active Power Control (ACTIVATE/ DEACTIVATE)	Double command (46)	25	1	Not used (0)	0	7
SCADA Reactive Power Control (ACTIVATE/ DEACTIVATE)	Double command (46)	26	1	Not used (0)	0	8

Outputs						
	Type ID	**IOA**	**Range**	**Interrogation Group**	**Cyclic Transmission Time**	**Index**
Output Voltage (KV)	Measured value, short floating point value (13)	500	1	Interrogated by station interrogation (20)	0	0
Output Active Power (MW)	Measured value, short floating point value (13)	501	1	Interrogated by station interrogation (20)	0	1
Output Reactive Power (MVar)	Measured value, short floating point value (13)	502	1	Interrogated by station interrogation (20)	0	2
Active SetPoint Feedback (MW)	Measured value, short floating point value (13)	503	1	Interrogated by station interrogation (20)	0	3
Reactive SetPoint Feedback (MVar)	Measured value, short floating point value (13)	504	1	Interrogated by station interrogation (20)	0	4
Plant Availability (Inverters/Total installed Inverters * 100%)	Measured value, short floating point value (13)	505	1	Interrogated by station interrogation (20)	0	5
Sun Shine (W/m2)	Measured value, short floating point value (13)	506	1	Interrogated by station interrogation (20)	0	6

Ait Temp. (C)	Measured value, short floating point value (13)	507	1	Interrogated by station interrogation (20)	0	7
Wind Speed (m/s)	Measured value, short floating point value (13)	508	1	Interrogated by station interrogation (20)	0	8
Power Factor (cos phi)	Measured value, short floating point value (13)	509	1	Interrogated by station interrogation (20)	0	9
Active Power Setpoint 100% feedback (On/Off)	Double-point information (3)	121	1	Interrogated by station interrogation (20)	0	10
Remote/Local Control of Main Circuit Breaker (0remote/1 local))	Single-point information (1)	101	1	Interrogated by station interrogation (20)	0	11
Local_SP_Override Feedback (Active/Not Active)	Single-point information (1)	102	1	Interrogated by station interrogation (20)	0	12
Active Power Setpoint new value (Accepted/Rejected)	Single-point information (1)	103	1	Interrogated by station interrogation (20)	0	13
Reactive Power Setpoint new value (Accepted/Rejected)	Single-point information (1)	104	1	Interrogated by station interrogation (20)	0	14
Overcurrent Protection (Operated/idle)	Single-point information (1)	105	1	Interrogated by station interrogation	0	15

				(20)		
Earth Fault Protection (Operated/idle)	Single-point information (1)	112	1	Interrogated by station interrogation (20)	0	16
Overfrequency Protection (Operated/idle)	Single-point information (1)	107	1	Interrogated by station interrogation (20)	0	17
Underfrequency Protection (Operated/idle)	Single-point information (1)	108	1	Interrogated by station interrogation (20)	0	18
Overvoltage Protection (Operated/idle)	Single-point information (1)	109	1	Interrogated by station interrogation (20)	0	19
Undervoltage Protection (Operated/idle)	Single-point information (1)	110	1	Interrogated by station interrogation (20)	0	20
Loss Of Main Protection (Operated/idle)	Single-point information (1)	111	1	Interrogated by station interrogation (20)	0	21
Grid Relay Malfunction (Operated/idle)	Single-point information (1)	106	1	Interrogated by station interrogation (20)	0	22
Circuit breaker Status (Opened/Closed)	Double-point information (3)	100	1	Interrogated by station interrogation (20)	0	23
Active Power Setpoint 60% feedback	Double-point information (3)	122	1	Interrogated by	0	24

(On/Off)				station interrogation (20)		
Active Power Setpoint 30% feedback (On/Off)	Double-point information (3)	123	1	Interrogated by station interrogation (20)	0	25
Active Power Setpoint 0% feedback (On/Off)	Double-point information (3)	124	1	Interrogated by station interrogation (20)	0	26
SCADA Active Power Control Feedback (ACTIVATED/DEACTIVATED)	Double-point information (3)	125	1	Interrogated by station interrogation (20)	0	27
SCADA Reactive Power Control Feedback (ACTIVATED/DEACTIVATED)	Double-point information (3)	126	1	Interrogated by station interrogation (20)	0	28

5.3 NETWORK CONFIGURATION

IP CONFIGURATION

IP ADDRESS	MASK	GATEWAY	EQUIPMENT	LOCATION
10.2.2.1	**255.255.255.0**	**10.2.2.254**	**Yaskawa - PLC**	**CABINET**
10.2.2.2	**255.255.255.0**	**10.2.2.3**	**IXXAT – IEC 104 GATEWAY**	**CABINET**
10.2.2.3	**255.255.255.0**	**10.2.2.254**	**Teltonika – GPRS router**	**CABINET**
10.2.2.10	**255.255.255.0**	**10.2.2.254**	**Smart Logger**	**CABINET**
10.2.2.11	**255.255.255.0**	**10.2.2.3**	**Siemens - Power Quality Meter**	**CABINET**
10.2.2.12	**255.255.255.0**	**10.2.2.254**	**Siemens - Protection Relay**	**CABINET**
10.2.2.30	**255.255.255.0**	**10.2.2.254**	**ICP/DAs – Modbus Gateway**	**CABINET**
10.2.2.100	**255.255.255.0**	**10.2.2.254**	**HP - Work Station**	**CABINET**
10.2.2.254	**255.255.255.0**	**10.2.2.254**	**Mikrotik - Router**	**CABINET**

5.4 EAC GPRS ROUTER

- Connection type QMI
- Mode NAT
- APN eacm2m
- Authentication method : PAP
- Username : Galascope2_PV
- Password : Galascope2PV_M2M
- Service mode : Automatic

PORTS FORWARDING

PROTOCOL	SOURCE	VIA	DESTINATION
TCP, UDP	**From any host in wan**	**To any router IP at port 80**	**Forward to IP 10.2.2.11 port 80 in lan**
TCP, UDP	**From any host in wan**	**To any router IP at port 102**	**Forward to IP 10.2.2.11 port 102 in lan**

5.5 Modbus Gateway

- Baud Rate (bps) 19200
- Data Size (bits) 8
- Parity Even
- Stop Bits (bits) 1
- Flow Control None
- Protocol RTU

5.6 Smart Logger

SN	Slave Address	FullName
2101074398ESK7001652	3	INV_1.01
2101074398ESK7000932	4	INV_1.02
2101074398ESK7001697	5	INV_1.03
2101074398ESK7000891	6	INV_1.04
2101074398ESK7000890	7	INV_1.05
2101074398ESK7001071	8	INV_1.09
2101074398ESK7000181	9	INV_2.03
2101074398ESK7000483	10	INV_1.07
2101074398ESK7001709	11	INV_2.11
2101074398ESK7001664	12	INV_1.11
2101074398ESK7001504	13	INV_2.06
2101074398ESK7000471	14	INV_2.02
2101074398ESK7001704	15	INV_2.05
2101074398ESK7000223	16	INV_2.01
2101074398ESK7000683	17	INV_2.08
2101074398ESK7000980	18	INV_1.08
2101074398ESK7000470	19	INV_1.06
2101074398ESK7000686	20	INV_2.04
2101074398ESK7001512	21	INV_1.10
2101074398ESK7000152	22	INV_2.09
2101074398ESK7000193	23	INV_2.10
2101074398ESK7001595	24	INV_2.07
EM00101950039322	246	EMI(COM1-246)
EM01101950039322	247	EMI(COM1-247)

5.7 Scada

Communication Protocols:

id	driver	Prefix	Address	Port	Modbus ID
prot1	Modbus TCP Client	PQM	10.2.2.11	502	
prot2	Modbus TCP Client	PR	10.2.2.12	502	
prot4	S7ET	PLC	10.2.2.1		
prot6	Modbus TCP Server				1

Calculation Routines:

Function Name	**Schedule Time**	**Action**	**Location**
Calculation_OnAction	Every 30 sec	Compute energy meter of inverters	
Export_OnAction	At 11h58	Export Data from trends (cumulative file)	C:/ Data/Trend_Energy/ YY_MM_DD.csv C:/ Data/ Trend_Inverter_Power/ YY_MM_DD.csv C:/ Data/ Trend_PQM/ YY_MM_DD.csv
Get_current_timedate	Every 30 sec		
ENABLE_AVAILABLE_OnAction	5 mn	Enable Alrm only if Sun Radiation is above value	
AliveEmailmorning	At 8h30	Send email with report as attachment and plant status	
AliveEmailnoon	At 13h30	Send email with plant status	
Send_report_daily	At 11h59	Generate daily report in pdf	C:/ Report/ report.pdf
Fileexport_OnAction	1Hr	Generate daily file with all inverters Energy values	C:/Source/ YY_MM_DD_Energy.csv
FileexportData_OnAction	2 mn	Generate daily file with: Ambient Temp PV Temp SunRadiation WindSpeedr Setpoint KW and KVAR from TSO Active power ReactivePower Frequency Power Factor Average voltage	C:/Source/ YY_MM_DD_Data.csv
TCU1_Alarm_OnAction	50 sec	Extract flag alarm of each TCUt	

Calculation Trend log:

There are 3 type of trends at the end of the day the file is saved on the disk

Trend_Inverter_Power

- Inverter_Active_Power
- Inverter_Reactive_power

Trend_PQM

- PQM_KW
- PQM_KVAR
- Inverter_Reactive
- PQM_AVERAGE_V
- PQM_PF
- PQM_HZ
- IEC_KW_SP
- IEC_KVAR_SP

Trend_Energy

- Inverter_Total_Energy
- Inverter Energy from 1 to 21

5.8 FTP Server /Web Interface

Some folders are created on the hard disk at location C: we have created the following folders:

C:\Alarms
C:\data
C:\report
C:\Source

Those folders are accessible via FTP client
Host : 185.230.112.161
User galascope
Password Galascope123
Port:2323

External IP : 185.230.112.161

With the local scada we can monitor the plant (Inverters/PQM/Protection Relay/Weather/EAC commands) This scada is web base and can be visualized from a web browser using the external IP and the port 85
User galascope
Password Galascope123

5.9 EMAILS STRUCTURE

Emails are send when an alarm is trigged or two times per day for reporting.

All alarms condition :

- Tracker TCUxx alarm
- Power Limitation PLant set at 0% by EAC
- Power Limitation PLant set at 30% by EAC
- Power Limitation PLant set at 60% by EAC
- Power Limitation PLant set at 100% by EAC
- Tracker TCU3 alarm
- Main CB Closed
- Main CB in Remote
- Main CB in Local
- Active Set point control on
- Plant Availability <100%
- UPS_By-Pass
- UPS_Failure
- Datalog Satus
- Main CB Opened
- TCU ALARMS DISABLED
- Protection OverCurrent
- Protection Earth Fault
- Protection OverVoltage
- Protection UnderVoltage
- Protection OverFrequency

- Protection UnderFRequency
- Protection RoCof
- Reactive Set point control on

Email structured for alarms:

Values between [xx] are dynamic

subject> Galascope 2.5MW : [Alarm Description]</subject>
<message> Galascope 2.5MW : [Alarm Description]
Plant Availability : [IEC_Availability] %

Inverter Status : [SL_STATUS] (***value between 0 and 7 is described at the end of the email)***
Active Power :[IEC_KW] KW
Reactive Power:[IEC_KVAR] KVar

Total Daily Energy:[Inverter_Daily_KWh] Kwh

Ambient Temp. : [AmbientTemp] °C
PV Temp. : [PVTemp] °C
Wind : [WindSpeed] m/s
Sun Radiation :[SUnRadiation] W/m²

Inverter Status Value : 0 IDLE
Inverter Status Value : 1 On-Grid
Inverter Status Value : 2 On-Grid : Self derating
Inverter Status Value : 3 On-Grid : Power limit
Inverter Status Value : 4 Planned Outage
Inverter Status Value : 5 Power limit outage
Inverter Status Value : 6 Fault outage
Inverter Status Value : 7 Communication interrupt</message>

Email structured for reports:

<subject> Galascope 2.5MW :Alive System</subject>
<message>Galascope 2.5MW :

Plant Availability : [IEC_Availability] %

Inverter Status : [SL_STATUS] (***value between 0 and 7 is described at the end of the email)***
Active Power :[IEC_KW] KW
Reactive Power:[IEC_KVAR] KVar

Total Daily Energy:[Inverter_Daily_KWh] Kwh

Ambient Temp. : [AmbientTemp] °C
PV Temp. : [PVTemp] °C
Wind : [WindSpeed] m/s
Sun Radiation :[SUnRadiation] W/m²

Inverter Status Value : 0 IDLE
Inverter Status Value : 1 On-Grid
Inverter Status Value : 2 On-Grid : Self derating
Inverter Status Value : 3 On-Grid : Power limit
Inverter Status Value : 4 Planned Outage
Inverter Status Value : 5 Power limit outage
Inverter Status Value : 6 Fault outage
Inverter Status Value : 7 Communication interrupt</message>